

IT & IS Audit Homework Assignment

Title: Data Breach Analysis through the Lens of IT Audit

Objective: The purpose of this assignment is to enhance your research and analytical skills by examining a recent data breach from a security and IT audit perspective. This exercise aims to develop your ability to identify security weaknesses, understand regulatory implications, and propose actionable improvements based on established audit frameworks.

Assignment Instructions

1. Choose a Data Breach (Individual Selection)

Select a publicly disclosed data breach that occurred within the last two years (2022-2024). Ensure that your chosen breach is well-documented in reputable sources such as news articles, regulatory reports, or company statements.

Examples (you may choose one or find your own):

- MOVEit Transfer breach (2023)
- Optus data breach (2022)
- Capita cyberattack (2023)
- MGM Resorts ransomware attack (2023)

2. Research and Document the Incident

Conduct research using at least three credible sources to gather details about the breach. Include references to your sources at the end of the assignment.

Analysis Framework

Use the following structured framework to guide your analysis:

Section 1: Overview of the Breach (10 points)

- What happened? Describe the breach (who, what, when, where). Include key facts such as the type of attack (e.g., ransomware, phishing, supply chain attack), the data compromised, and the impact on the organization and affected parties.
- Timeline: Provide a timeline of the breach from detection to public disclosure.

Section 2: Security Failures and Vulnerabilities (20 points)

- Root cause analysis: Identify what security controls were likely missing or inadequate.

- Audit relevance: How could an IT audit have identified these vulnerabilities before the breach occurred?

Section 3: Regulatory and Legal Implications (15 points)

- Regulations impacted: Identify any data protection laws or industry standards that the organization likely violated (e.g., GDPR, CCPA, PCI DSS).
- Consequences: Highlight any fines, lawsuits, or reputational damage that resulted from the breach.

Section 4: Recommendations and Lessons Learned (20 points)

- Proposed improvements: Suggest three security measures the organization should implement to prevent future breaches.
- Audit frameworks: Explain how IT audit frameworks (such as COBIT, ISO 27001, or NIST) could have helped mitigate the risks.

Section 5: Personal Reflection & Critical Thinking (25 points)

- Application to IT audit: Reflect on how this breach has influenced your understanding of IT & IS audit practices. Consider how you would apply the lessons learned to an IT audit in your own (real or hypothetical) organization.
- Critical Thinking Statement: Write a short paragraph (150-200 words) summarizing what you personally found most surprising, challenging, or insightful during your research. Highlight any difficulties you faced in gathering information and how this assignment has shaped your understanding of cybersecurity and audit practices.
- Imagine you are the IT auditor responsible for this organization a year before the breach. What red flags would you look for during your audit, and what recommendations would you make to prevent this breach?

Submission Requirements

- Length: 1,500 - 2,000 words.
- Formatting: Use a clear, professional format with headings for each section.
- References: Include a bibliography citing all sources used.

Grading Criteria

Section	Points
Overview of the Breach	10
Security Failures	20

Section	Points
Regulatory Implications	20
Recommendations	20
Personal Reflection & Critical Thinking	30
Total	100

Tips for Success

1. Avoid copy-pasting content from your sources. Provide original analysis and insights.
2. Use multiple sources to get a well-rounded understanding of the incident.
3. Cite your references properly to avoid plagiarism.
4. Apply critical thinking and connect your findings to IT audit principles.